Example 6.15

Calculate the consumer price index number for 2015 on the basis of 2000 from the following data by using (i) the Aggregate expenditure method (ii) the family budget (or) weighted relatives method.

Commodity	Quantity	Price	
		2000	2015
Wheat	20	15	20
Rice	8	20	24
Ghee	2	160	200
Sugar	4	40	40

Solution

(i) Calculation of cost of living index number on the basis of Aggregate expenditure method.

Commodity	q_0	p_0	p_1	p_0q_0	p_1q_0
Wheat	20	15	20	300	400
Rice	8	20	24	160	192
Ghee	2	160	200	320	400
Sugar	4	40	40	160	160
Total				940	1152

Consumer price index number for 2015

$$P_{01} = \frac{\sum p_1q_0}{\sum p_0q_0} \times 100$$

$$= \frac{1152}{940} \times 100$$

$$\approx 112.6$$

(iii) Calculation of consumer price index number according to family budget method or weighted relative method

Commodity	q_0	p_0	p_1	$p = \frac{p_1}{p_0} \times 100$	$w = p_0q_0$	wp
Wheat	20	15	20	400/3	300	40000
Rice	8	20	24	120	160	19200
Sugar	2	160	200	125	320	40000
Ghee	4	40	40	100	160	16000
					940	115200

Consumer price index number for 2015

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